

Unified Off-Campus Recruitment System & Automated ATS Engine

A End-to-End Platform for Candidate Pool Control, In-App Assessment, WebRTC Interviews, and Recruiter Decision SLAs

1. Executive Summary

The modern off-campus recruitment ecosystem is broken for both job seekers and hiring managers. Job boards like LinkedIn and Naukri operate primarily as passive document repositories. When a company posts an entry-level or junior role, recruiters are inundated with thousands of unvetted, irrelevant applications. Consequently, recruiters spend excessive manual effort filtering resumes, leading to high candidate drop-off rates and systemic candidate ghosting—where applicants receive rejection notices months late or no feedback at all.

This project proposes **HireSync** (or **TalentPulse ATS**), an end-to-end, automated off-campus recruitment platform. The system combines strict eligibility gatekeeping, an automated Applicant Tracking System (ATS) parsing engine, in-app skill evaluations, embedded video interviewing capabilities, and an automated Service-Level Agreement (SLA) decision pipeline. By bringing all hiring stages under a single web ecosystem, the platform eliminates candidate ghosting and significantly reduces recruitment overhead for companies.

2. Problem Statement & Industry Gap

2.1 Candidate Ghosting and Delayed Feedback

In traditional hiring workflows, candidates submit applications into an unmonitored pipeline. Because recruiters lack automated tracking and decision enforcement mechanisms, candidates are left in "recruitment limbo." Rejection emails are frequently triggered 3 to 6 months after the initial application, degrading the candidate experience and creating mistrust in off-campus recruitment channels.

2.2 Unfiltered Application Volume (Recruiter Fatigue)

When a job posting does not enforce strict upstream eligibility checks, thousands of unqualified applicants submit resumes. Manual resume evaluation is inefficient, error-prone, and unsustainable for small-to-medium enterprises (SMEs) and early-stage startups that lack large Human Resources (HR) departments.

2.3 Fragmented Hiring Tools

Currently, the off-campus hiring process spans multiple disconnected tools:

- Resumes are collected via email or job boards.
- Assessments are conducted using external forms or third-party coding portals.
- Interviews are scheduled through third-party video conferencing links.
- Candidate tracking is managed manually in static spreadsheets.

This fragmentation leads to data loss, scheduling friction, and a lack of operational transparency.

3. Proposed Solution & Core Features

The proposed platform acts as an integrated middleware connecting candidates and recruiters through five core architectural pillars:

3.1 Upstream Eligibility Gatekeeper

- **Hard Filters:** Restricts job visibility and locks application buttons based on predefined criteria such as total years of experience, specific graduation batch years, degree streams, and maximum notice period.
- **Skill-Based Unlock Mechanism:** Allows candidates who fall slightly below experience thresholds (e.g., 1.5 years instead of 2 years) to attempt a timed, automated skill assessment to unlock their application.

3.2 Automated ATS Engine & Resume Parser

- **PDF Document Parsing:** Extracts structured data (work history, skills, education, contact details) from uploaded PDF resumes.
- **Algorithmic Match Scoring:** Calculates a weighted percentage match score based on job description requirements versus candidate profile metrics.
- **Automated Candidate Tiering:** Categorizes applicants automatically into Tier-1 (Ideal Match), Tier-2 (Conditional / Needs Assessment), and Tier-3 (Ineligible) to streamline HR review.

3.3 Integrated In-App Assessment & WebRTC Interviewing

- **Native Testing Engine:** Auto-graded multiple-choice and short-form skill tests are served directly within the user dashboard.
- **Embedded Video Interviewing:** Built-in WebRTC video rooms allow recruiters to conduct live interviews with real-time evaluation scorecards embedded next to the video frame, eliminating external meeting links.

3.4 Live Application Status Tracking

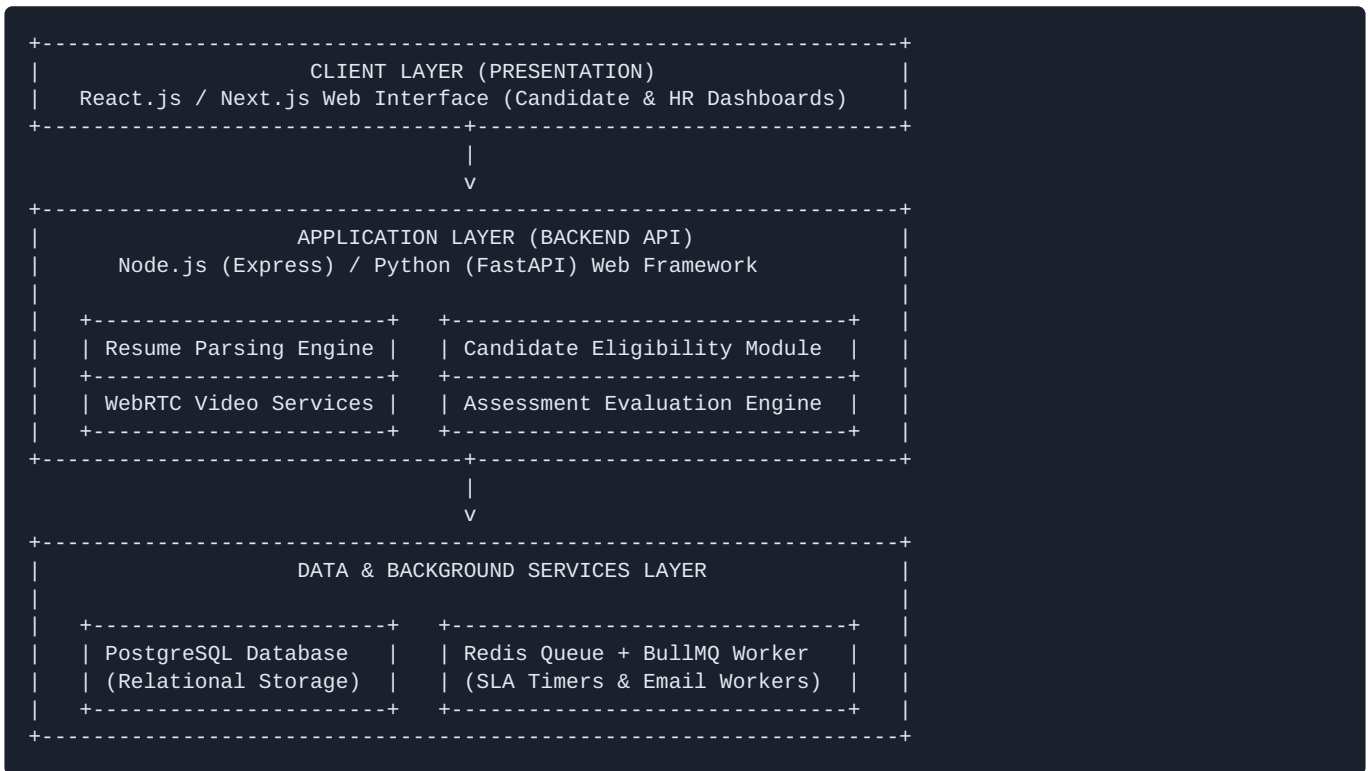
Provides candidates with a real-time, stage-by-stage pipeline tracker (*Applied* → *Screened* → *Assessment* → *Interview* → *Decision*) with maximum estimated waiting times for each stage to maintain complete transparency.

3.5 Recruiter Decision SLA Engine

Implements a background timer (e.g., 7 days) upon the completion of a candidate's assessment or interview. It triggers automated alerts to recruiters as the deadline approaches and executes automated fallback actions (such as releasing polite, automated rejection/feedback emails) if the recruiter fails to take manual action before timer expiration.

4. System Architecture & Data Flow

The platform is structured using a three-tier web application architecture:



Data Flow Sequence:

1. **Recruiter** posts a job with minimum eligibility criteria, evaluation weightages, and SLA window.
2. **Candidate** accesses the portal. The system filters available job listings based on candidate profile data.
3. If eligible, candidate applies by submitting a resume. The **ATS Engine** parses the file and generates a match index.
4. Candidates meeting Tier-1 or Tier-2 criteria are automatically assigned an in-app assessment.
5. Upon passing, an in-app WebRTC video interview slot is scheduled.
6. The background **SLA Worker** initiates a countdown clock. If HR evaluates the interview, the status updates immediately. If inactive upon timeout, the system executes automated candidate notification routines.

5. Technical Stack Specification

- **Frontend Framework:** `React.js / Next.js` `Tailwind CSS` (State management, server-side rendering, and responsive UI components).
- **Backend Framework:** `Node.js (Express)` OR `Python (FastAPI)` (REST API endpoint design and request routing).
- **Database:** `PostgreSQL` (Relational schema modeling for Users, Jobs, Applications, Test Scores, and Interview Logs).
- **In-Memory Cache & Job Queue:** `Redis` + `BullMQ / Celery` (Handling background timer tasks, SLA countdowns, and automated email notifications).
- **Real-Time Communication:** `WebRTC API / Agora SDK` (Peer-to-peer browser-based video calls).
- **Parsing & Utilities:** `pdfplumber / pdf-parse` and `Nodemailer / SendGrid` (Document text extraction and transactional email delivery).

6. Competitive Comparison

Feature / Metric	Traditional Job Portals (e.g., Naukri)	Native ATS Systems (e.g., Greenhouse)	Proposed Unified Platform
Candidate Screening	Post-apply manual filtering	Manual resume review by HR	Upstream eligibility locking + Automated ATS parsing
Tool Integration	Third-party redirects (external forms/links)	Separate ATS instance	Fully unified (Assessments, Video, ATS in one portal)
Recruiter SLA Enforcement	None (applications stay open indefinitely)	Manual notification reminders	System-enforced countdown timers with automated fallbacks
Candidate Feedback	High rate of ghosting / delayed responses	Dependent on HR manual execution	Guaranteed live status tracking and automated SLA updates

7. Project Roadmap & Milestones

- **Phase 1: Requirement Analysis & Database Schema Design (Week 1)**
Finalize product requirements document (PRD), design relational database schemas, and set up repositories.
- **Phase 2: Core Platform Development & Gatekeeping (Week 2)**
Develop user authentication (JWT-based role management), job posting modules with restriction logic, and Candidate Dashboard.
- **Phase 3: ATS Engine, Assessment, & SLA Integration (Week 3)**
Integrate PDF resume parsing algorithms, build in-app assessment module, and configure Redis background queues for SLA timers.
- **Phase 4: WebRTC Video Call, Testing, & Deployment (Week 4)**
Implement embedded WebRTC video call rooms, conduct end-to-end integration testing, and deploy to cloud hosting services.

8. Expected Outcomes & Academic Value

1. **Reduction in Manual HR Labor:** Demonstrates how automated gatekeeping and resume parsing can reduce recruiter evaluation time by up to 90%.
2. **Elimination of Candidate Ghosting:** Proves that queue-based SLA systems can guarantee timely application updates to candidates.
3. **Engineering Depth:** Demonstrates advanced computer science competencies including asynchronous job processing, background queuing systems, document parsing algorithms, relational database design, and real-time media transport protocols.